

Physics 513 QFT HW 1

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CONTENTS

1. 1

2. 2

2.1. a

2.2. b

3. 3

3.1. a

3.2. b

1. 1

I use Cartesian coordinates

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z). \quad (1)$$

I use matrix (not component) notation. The rotation operator around the x axis is

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \phi & \sin \phi \\ 0 & 0 & -\sin \phi & \cos \phi \end{pmatrix}, \quad (2)$$

For an infinitesimal rotational angle $|\delta\phi| \ll 1$

$$R_x(\delta\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \delta\phi \\ 0 & 0 & -\delta\phi & 1 \end{pmatrix}. \quad (3)$$

$$R_x(\delta\phi) = \mathbf{I} - \delta\phi J_x, \quad (4)$$

where $\mathbf{I}$ is the identity matrix and

$$J_x = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (5)$$

Rotations about the x axis obey the composition property: $R_x(\phi)R_x(\theta) = R_x(\phi + \theta)$. Then a finite rotation can be compounded from an infinite number of infinitesimal rotations $\delta\phi = \phi/N$:

$$R_x(\phi) = \lim_{N \rightarrow \infty} R_x(\delta\phi)^N = \lim_{N \rightarrow \infty} (\mathbf{I} - \delta\phi J_x)^N = \lim_{N \rightarrow \infty} \left(\mathbf{I} - \frac{\phi}{N} J_x \right)^N. \quad (6)$$

There is a neat identity

$$e^{-yx} = \lim_{N \rightarrow \infty} \left(1 - \frac{y}{N}x \right)^N, \quad (7)$$

that generalizes for matrices

$$R_x(\phi) = e^{-\phi J_x} = e^{i^2 \phi J_x} = e^{i\phi J_x} = e^{i\phi L_x}. \quad (8)$$

In group theory jargon,

$$\boxed{L_x = iJ_x} \quad (9)$$

is the generator of R_x .

I repeat this exercise for rotations around the y and z axes.

$$R_y(\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & 0 & -\sin \phi \\ 0 & 0 & 1 & 0 \\ 0 & \sin \phi & 0 & \cos \phi \end{pmatrix} \quad (10)$$

$$R_y(\delta\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\delta\phi \\ 0 & 0 & 1 & 0 \\ 0 & \delta\phi & 0 & 1 \end{pmatrix} = \mathbf{I} - \delta\phi J_y \quad (11)$$

$$\boxed{J_y = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}} \quad (12)$$

$$R_y(\phi) = \lim_{N \rightarrow \infty} R_y(\delta\phi)^N = \lim_{N \rightarrow \infty} R_y \left(\frac{\phi}{N} \right)^N =$$

$$\lim_{N \rightarrow \infty} \left(\mathbf{I} - \frac{\phi}{N} J_y \right)^N = e^{-\phi J_y} = e^{i^2 \phi J_y} = e^{i\phi J_y} = e^{i\phi L_y} \quad (13)$$

$$\boxed{L_y = iJ_y} \quad (14)$$

$$R_z(\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & \sin \phi & 0 \\ 0 & -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (15)$$

$$R_z(\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & \delta\phi & 0 \\ 0 & -\delta\phi & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{I} - \delta\phi J_z \quad (16)$$

$$J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (17)$$

$$R_z(\phi) = e^{i\phi L_z}, \quad \boxed{L_z = iJ_z} \quad (18)$$

The boost along the x direction is

$$B_x(\alpha) = \begin{pmatrix} \cosh \alpha & \sinh \alpha & 0 & 0 \\ \sinh \alpha & \cosh \alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (19)$$

where α is the rapidity parameter. The boost must be written in terms of the rapidity parameter, and not the velocity, because the rapidity parameter obeys the composition rule, but the velocity does not

$$B_x(\alpha)B_x(\beta) = B_x(\alpha + \beta). \quad (20)$$

For an infinitesimal rapidity parameter

$$\cosh \delta\alpha = 1 + \mathcal{O}(\delta\alpha^2), \quad \sinh \delta\alpha \approx \delta\alpha + \mathcal{O}(\delta\alpha^3), \quad (21)$$

$$B_x(\delta\alpha) = \begin{pmatrix} 1 & \delta\alpha & 0 & 0 \\ \delta\alpha & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{I} - \delta\alpha V_x, \quad (22)$$

$$V_x = \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (23)$$

I proceed like I did with rotations but with $\delta\alpha = \alpha/N$,

$$\begin{aligned} B_x(\alpha) &= \lim_{N \rightarrow \infty} B_x(\delta\alpha)^N = \lim_{N \rightarrow \infty} B_x\left(\frac{\alpha}{N}\right)^N = \\ \lim_{N \rightarrow \infty} \left(\mathbf{I} - \frac{\alpha}{N} V_x\right)^N &= e^{-\alpha V_x} = e^{i^2 \alpha V_x} = e^{i\alpha i V_x} = e^{i\alpha K_x} \end{aligned} \quad (24)$$

and the generator of B_x is

$$\boxed{K_x = iV_x}. \quad (25)$$

I repeat the calculation for boosts along the y and z directions.

$$B_y(\alpha) = \begin{pmatrix} \cosh \alpha & 0 & \sinh \alpha & 0 \\ 0 & 1 & 0 & 0 \\ \sinh \alpha & 0 & \cosh \alpha & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (26)$$

$$B_y(\delta\alpha) = \begin{pmatrix} 1 & 0 & \delta\alpha & 0 \\ 0 & 1 & 0 & 0 \\ \delta\alpha & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{I} - \delta\alpha V_y \quad (27)$$

$$V_y = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (28)$$

$$B_y(\alpha) = e^{i\alpha K_y}, \quad \boxed{K_y = iV_y} \quad (29)$$

$$B_z(\alpha) = \begin{pmatrix} \cosh \alpha & 0 & 0 & \sinh \alpha \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \alpha & 0 & 0 & \cosh \alpha \end{pmatrix} \quad (30)$$

$$B_z(\delta\alpha) = \begin{pmatrix} 1 & 0 & 0 & \delta\alpha \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \delta\alpha & 0 & 0 & 1 \end{pmatrix} = \mathbf{I} - \delta\alpha V_z \quad (31)$$

$$V_z = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \quad (32)$$

$$B_z(\alpha) = e^{i\alpha K_z}, \quad \boxed{K_z = iV_z} \quad (33)$$

These are the generators of the Lorentz group. They satisfy

$$\begin{aligned} [L_i, L_j] &= i\epsilon_{ijk} L_k, & [L_i, K_j] &= i\epsilon_{ijk} K_k, \\ [K_i, K_j] &= -i\epsilon_{ijk} L_k, \end{aligned} \quad (34)$$

where ϵ_{ijk} is the Levi-Civita symbol and there is no summation over repeated indices. I plugged the generators into these commutators in Mathematica and confirmed that they are true. QED.

2. 2

Without loss of generality I can restrict myself to 1 spatial dimension. My coordinates and metric are

$$x^\mu = (x^0, x^1) = (t, x), \quad \eta_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (35)$$

The two events are

$$x_0^\mu = (t_0, x_0), \quad x_1^\mu = (t_1, x_1). \quad (36)$$

The interval between them is

$$\Delta s^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu = \Delta t^2 - \Delta x^2, \quad (37)$$

where

$$\Delta t = t_1 - t_0, \quad \Delta x = x_1 - x_0. \quad (38)$$

The boost is

$$\Lambda_{\mu}^{\bar{\mu}} = \begin{pmatrix} \gamma & -\gamma v \\ -\gamma v & \gamma \end{pmatrix}, \quad (39)$$

where v is the boost velocity and

$$\gamma = \frac{1}{\sqrt{1-v^2}}. \quad (40)$$

The boosted coordinates are

$$x^{\bar{\mu}} = (x^{\bar{0}}, x^{\bar{1}}) = (\bar{t}, \bar{x}), \quad x^{\bar{\mu}} = \Lambda_{\mu}^{\bar{\mu}} x^{\mu}, \quad (41)$$

where

$$\bar{t} = x^{\bar{0}} = \Lambda_{\mu}^{\bar{0}} x^{\mu} = \gamma t - \gamma v x, \quad (42)$$

$$\bar{x} = x^{\bar{1}} = \Lambda_{\mu}^{\bar{1}} x^{\mu} = -\gamma v t + \gamma x. \quad (43)$$

The spacetime points, Eq. 36, in the boosted coordinates are

$$\begin{aligned} x_0^{\bar{\mu}} &= (\bar{t}_0, \bar{x}_0) = \gamma(t_0 - vx_0, x_0 - vt_0), \\ x_1^{\bar{\mu}} &= (\bar{t}_1, \bar{x}_1) = \gamma(t_1 - vx_1, x_1 - vt_1). \end{aligned} \quad (44)$$

The metric is invariant under boosts

$$\eta_{\bar{\mu}\bar{\nu}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (45)$$

so

$$\Delta s^2 = \Delta \bar{t}^2 - \Delta \bar{x}^2, \quad \Delta \bar{t} = \bar{t}_1 - \bar{t}_0, \quad \Delta \bar{x} = \bar{x}_1 - \bar{x}_0. \quad (46)$$

2.1. a

The interval is Lorentz invariant:

$$\Delta s^2 = \Delta t^2 - \Delta x^2 = \Delta \bar{t}^2 - \Delta \bar{x}^2. \quad (47)$$

The interval is spacelike

$$\Delta s^2 < 0. \quad (48)$$

If the events are simultaneous in the boosted frame I get the expression

$$\boxed{\Delta s^2 = -\Delta \bar{x}^2 < 0 \implies \text{TRUE}}. \quad (49)$$

This expression is self-consistent: therefore it is possible to find a Lorentz frame where spacelike separated events 1 & 2 are simultaneous. If they occur at the same spatial point in the boosted frame $\Delta \bar{x} = 0$:

$$\boxed{\Delta s^2 = \Delta \bar{t}^2 < 0 \implies \text{FALSE}}. \quad (50)$$

Spacelike separated events 1 & 2 cannot occur at the same spatial point in any frame. QED.

2.2. b

Now I assume the interval is timelike

$$\Delta s^2 > 0. \quad (51)$$

Using my notation from Sec. 2.1:

$$\boxed{\Delta s^2 = -\Delta \bar{x}^2 > 0 \implies \text{FALSE}}, \quad (52)$$

$$\boxed{\Delta s^2 = \Delta \bar{t}^2 > 0 \implies \text{TRUE}},$$

QED.

3. 3

My coordinates are

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z) = (t, \mathbf{r}). \quad (53)$$

In GR its not kosher to treat coordinates as vectors, but it's okay in SR. The 4-current field is

$$j^\mu = j^\mu(x^\nu) = j^\mu(t, \mathbf{r}), \quad j^\mu = (j^0, j^1, j^2, j^3) = (\rho, \mathbf{j}), \quad (54)$$

where

$$j^0 = \rho = \rho(t, \mathbf{r}), \quad (55)$$

is the charge density.

3.1. a

$$Q = \int d^3x j^0(t, \mathbf{r}), d^3x = dx dy dz = dV \quad (56)$$

$$Q = \int dV \rho(t, \mathbf{r}) \quad (57)$$

$$\dot{Q} = \frac{dQ}{dt} = \frac{d}{dt} \int dV \rho(t, \mathbf{r}) \quad (58)$$

$$\dot{Q} = \int dV \frac{d}{dt} \rho(t, \mathbf{r}) = \int dV \dot{\rho}(t, \mathbf{r}) \quad (59)$$

In this case there is no difference between the total and partial time derivatives.

$$\begin{aligned} \partial_\mu j^\mu &= 0 = \partial_0 j^0 + \partial_1 j^1 + \partial_2 j^2 + \partial_3 j^3 = \\ \partial_t j^t + \partial_x j^x + \partial_y j^y + \partial_z j^z &= \partial_t \rho + \nabla \cdot \mathbf{j} = \\ \dot{\rho} + \nabla \cdot \mathbf{j} &= 0 \implies \boxed{\dot{\rho}(t, \mathbf{r}) = -\nabla \cdot \mathbf{j}(t, \mathbf{r})}. \end{aligned} \quad (60)$$

$$\dot{Q} = - \int dV \nabla \cdot \mathbf{j}(t, \mathbf{r}) \quad (61)$$

I use the divergence theorem

$$\int dV \nabla \cdot \mathbf{j}(t, \mathbf{r}) = \oint dA \mathbf{j}(t, \mathbf{r}) \cdot \hat{\mathbf{n}}, \quad (62)$$

where my Gaussian surface is a sphere at infinity with unit normal $\hat{\mathbf{n}} = \hat{\mathbf{r}}$. The exact form of its infinitesimal area element dA is irrelevant.

$$\dot{Q} = - \oint dA \mathbf{j}(t, \infty) \cdot \hat{\mathbf{r}} \quad (63)$$

I make the reasonable assumption that there is no matter (so no charges and no current) at infinity $\mathbf{j}(t, \infty) = 0$.

$$\dot{Q} = - \oint dA \mathbf{j}(t, \infty) \cdot \hat{\mathbf{r}} = 0, \implies \boxed{\dot{Q} = 0} \quad (64)$$

QED.

3.2. b

The Lorentz group includes rotations and boosts. It is well known that charge is conserved under rotations, so I will only consider boosts. Because space is isotropic I lose no generality by only considering boosts in the x direction

$$\Lambda_{\mu}^{\bar{\mu}} = \begin{pmatrix} \gamma & -v\gamma & 0 & 0 \\ -v\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (65)$$

where v is the boost 3-velocity and

$$\gamma = \frac{1}{\sqrt{1-v^2}}. \quad (66)$$

My boosted coordinates are

$$x^{\bar{\mu}} = (x^{\bar{0}}, x^{\bar{1}}, x^{\bar{2}}, x^{\bar{3}}) = (\bar{t}, \bar{x}, \bar{y}, \bar{z}). \quad (67)$$

The charge in my original frame is

$$Q = \int d^3x \rho(t, x, y, z), \quad d^3x = dx dy dz. \quad (68)$$

The charge in the boosted frame is

$$\bar{Q} = \int d^3\bar{x} \bar{\rho}(\bar{t}, \bar{x}, \bar{y}, \bar{z}), \quad d^3\bar{x} = d\bar{x} d\bar{y} d\bar{z}. \quad (69)$$

There is a nice trick to solve this problem that I found in the Problem Book in Relativity and Gravitation by Lightman et al. (Problem 3.35). The integrals in Eqs. 68 & 69 are evaluated on 3D hypersurfaces of constant time ($t = t_0$ & $\bar{t} = \bar{t}_0$). These surfaces intersect and create two hypervolumes, R_1 & R_2 , (FIG. 1). I consider the integral

$$\int_{R_1} d^4x \partial_{\mu} J^{\mu} = \int_{R_1} dt dx dy dz \partial_{\mu} J^{\mu}(t, x, y, z), \quad (70)$$

over the spacetime hypervolume R_1 . The divergence theorem says

$$\int_{R_1} d^4x \partial_{\mu} J^{\mu} = \int_{S_1} d^3x J^{\mu} n_{\mu}, \quad (71)$$

where the hypersurface S_1 encloses the hypervolume R_1 , d^3x is its volume element, and n^{μ} is its outward pointing unit normal. Because $\partial_{\mu} J^{\mu} = 0$

$$\int_{R_1} d^4x \partial_{\mu} J^{\mu} = 0, \quad (72)$$

$$\oint_{S_1} d^3x J^{\mu} n_{\mu} = 0. \quad (73)$$

This integral can be broken up into several pieces

$$\begin{aligned} 0 &= \oint_{S_1} d^3x J^{\mu} n_{\mu} = \int_{\mathcal{X}} d^3x J^{\mu}(t, -\infty, y, z) n_{\mu} + \\ &\int_{\mathcal{Y}_-} d^3x J^{\mu}(t, x, -\infty, z) n_{\mu} + \int_{\mathcal{Y}_+} d^3x J^{\mu}(t, x, \infty, z) n_{\mu} + \\ &\int_{\mathcal{Z}_-} d^3x J^{\mu}(t, x, y, -\infty) n_{\mu} + \int_{\mathcal{Z}_+} d^3x J^{\mu}(t, x, y, \infty) n_{\mu} + \\ &\int_{\sigma_1(t_0)} d^3x J^{\mu} n_{\mu} + \int_{\sigma_1(\bar{t}_0)} d^3x J^{\mu} n_{\mu}, \end{aligned} \quad (74)$$

where $\mathcal{X}$ is the surface at $x = -\infty$, $\mathcal{Y}_{\pm}$ are surfaces at $y = \pm\infty$, $\mathcal{Z}_{\pm}$ are surfaces at $z = \pm\infty$, $\sigma_1(t_0)$ is the surface of constant time $t = t_0$, and $\sigma_1(\bar{t}_0)$ is the surface of constant boosted time $\bar{t} = \bar{t}_0$. The red and blue colors match the colors in FIG. 1. But there is no matter at infinity so

$$J^{\mu}(t, \pm\infty, y, z) = J^{\mu}(t, x, \pm\infty, z) = J^{\mu}(t, x, y, \pm\infty) = 0, \quad (75)$$

$$\begin{aligned} 0 &= \int_{\mathcal{X}} d^3x J^{\mu}(t, -\infty, y, z) n_{\mu} + \\ &\int_{\mathcal{Y}_-} d^3x J^{\mu}(t, x, -\infty, z) n_{\mu} + \int_{\mathcal{Y}_+} d^3x J^{\mu}(t, x, \infty, z) n_{\mu} + \\ &\int_{\mathcal{Z}_-} d^3x J^{\mu}(t, x, y, -\infty) n_{\mu} + \int_{\mathcal{Z}_+} d^3x J^{\mu}(t, x, y, \infty) n_{\mu} + \\ &\int_{\sigma_1(t_0)} d^3x J^{\mu} n_{\mu} + \int_{\sigma_1(\bar{t}_0)} d^3x J^{\mu} n_{\mu}, \end{aligned} \quad (76)$$

$$0 = \int_{\sigma_1(t_0)} d^3x J^{\mu} n_{\mu} + \int_{\sigma_1(\bar{t}_0)} d^3x J^{\mu} n_{\mu}. \quad (77)$$

The outward unit normal along the $t = t_0$ surface is

$$n^{\mu} = (1, 0, 0, 0). \quad (78)$$

I lower the index $n_\mu = \eta_{\mu\nu} n^\nu$, where $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$, so

$$\eta_\mu = (1, 0, 0, 0). \quad (79)$$

Note that $n^\mu n_\mu = 1$ so n^μ really is the *unit* normal. Then

$$\begin{aligned} \int_{\sigma_1(t_0)} d^3x J^\mu n_\mu &= \int_{\sigma_1(t_0)} d^3x J^0 n_0 = \int_{\sigma_1(t_0)} d^3x J^0 = \\ &= \int_{\sigma_1(t_0)} d^3x \rho. \end{aligned} \quad (80)$$

To work out the blue integral it is easier to work in the boosted frame. But because the integrand is a scalar that's easy.

$$n^{\bar{\mu}} = n_{\bar{\mu}} = (-1, 0, 0, 0), \quad (81)$$

The minus sign appears because I need the *outward* unit normal. Then

$$J^\mu n_\mu = J^{\bar{\mu}} n_{\bar{\mu}} = J^{\bar{0}} n_{\bar{0}} = -J^{\bar{0}} = -\bar{\rho}, \quad (82)$$

$$\begin{aligned} \int_{\sigma_1(\bar{t}_0)} d^3x J^\mu n_\mu &= - \int_{\sigma_1(\bar{t}_0)} d^3x \bar{\rho} = \\ &= - \int_{\sigma_1(\bar{t}_0)} dx dy dz \bar{\rho}(t, x, y, z) = \\ &= - \int_{\sigma_1(\bar{t}_0)} d\bar{x} d\bar{y} d\bar{z} \bar{\rho}(\bar{t}_0, \bar{x}, \bar{y}, \bar{z}) = - \int_{\sigma_1(\bar{t}_0)} d^3\bar{x} \bar{\rho} \end{aligned} \quad (83)$$

The t in the integral above is constrained by the bounds of integration: I am integrating over the hypersurface $\sigma_1(\bar{t}_0)$. The change of coordinates is allowed because I can integrate the function $\bar{\rho}$ in whatever coordinates I want. Plugging Eqs. 80 & 83 into 77 yields

$$\int_{\sigma_1(\bar{t}_0)} d^3\bar{x} \bar{\rho} = \int_{\sigma_1(t_0)} d^3x \rho. \quad (84)$$

Repeating this calculation for R_2 yields

$$\int_{\sigma_2(\bar{t}_0)} d^3\bar{x} \bar{\rho} = \int_{\sigma_2(t_0)} d^3x \rho, \quad (85)$$

where the σ_2 refers to the boundary surface of R_2 . Combining Eqs. 84 & 85 yields

$$\int_{\sigma_1(\bar{t}_0)} d^3\bar{x} \bar{\rho} + \int_{\sigma_2(\bar{t}_0)} d^3\bar{x} \bar{\rho} = \int_{\sigma_1(t_0)} d^3x \rho + \int_{\sigma_2(t_0)} d^3x \rho, \quad (86)$$

$$\int_{\sigma_1(\bar{t}_0) + \sigma_2(\bar{t}_0)} d^3\bar{x} \bar{\rho} = \int_{\sigma_1(t_0) + \sigma_2(t_0)} d^3x \rho, \quad (87)$$

where $\sigma_1(t_0) + \sigma_2(t_0)$ is the entire $t = t_0$ hypersurface.

$$\int d^3\bar{x} \bar{\rho} = \int d^3x \rho \quad (88)$$

Comparing to Eqs. 68 & 69 yields

$$\bar{Q} = Q, \quad (89)$$

QED.

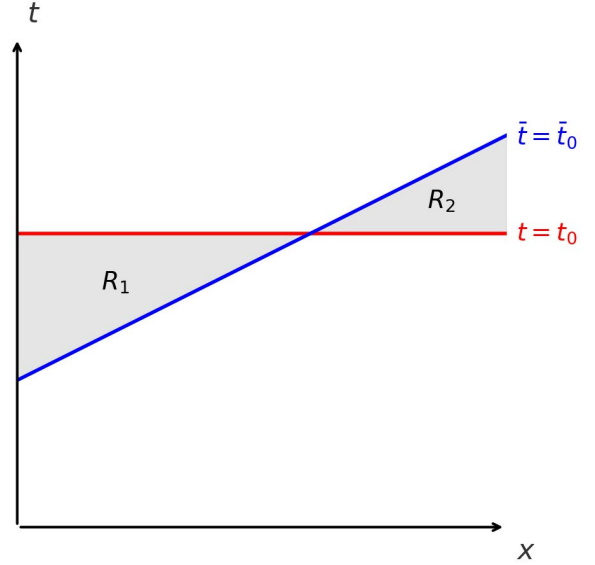


FIG. 1. Spatial hypersurfaces of constant time along which I integrate Q (red) and $\bar{Q}$ (blue). The R_n are the spacetime hypervolumes I exploit with the divergence theorem. I suppressed the y and z directions.